

PRACTICAL NO 6 MODULE 2

AIM: Paired t-test

OUTPUT:-

The screenshot displays the RStudio interface with the following components:

- Source Editor:** Contains R code for loading the 'airline_flight_dataset.csv' file, creating a data frame 'df', and performing a paired t-test on 'df\$delay_before' and 'df\$delay_after'.
- Console:** Shows the output of the R code, including the structure of the data frame and the results of the paired t-test.
- Environment:** Lists the objects in the global environment, including 'airline_flight_d...', 'df', 'ecommerce_sales...', 'professional_hr...', 't_test_one', 't_test_paired', and 't_test_two'.
- Files:** Shows the file explorer with various files and folders, including 'PRACTICAL NO 8.R', 'PRACTICAL NO 9.R', 'PRACTICAL NO 10.R', 'Processed_Global_Mobile_Prices.csv', 'SALARY.xlsx', 'synthetic_freelance_jobs.csv', 'uber_driver_trip_analysis.csv', and several WhatsApp images.

```
<R> R 4.5.2 - C:/Users/as993/DATA SET />
> df <- read.csv("airline_flight_dataset.csv")
> str(df)
'data.frame':   7000 obs. of  10 variables:
 $ Flight_ID      : int  500001 500002 500003 500004 500005 500006 500007 500008 500009 500010 ...
 $ Airline        : chr   "Air India" "IndiGo" "Akasa Air" "IndiGo" ...
 $ Source_City    : chr   "Hyderabad" "Mumbai" "Bengaluru" "Hyderabad" ...
 $ Destination_City : chr   "Hyderabad" "Bengaluru" "Delhi" "Kolkata" ...
 $ Flight_Distance_km : int  630 553 1900 1503 755 1480 2169 2343 1189 1998 ...
 $ Scheduled_Departure: chr   "2023-06-01 00:00:00" "2023-06-01 01:00:00" "2023-06-01 02:00:00" "2023-06-01 03:00:00" ...
 $ Departure_Delay_Min: int   109 -9 32 0 161 37 57 90 104 109 ...
 $ Arrival_Delay_Min : int   60 131 143 21 57 95 49 79 42 127 ...
 $ Ticket_Price     : int  11406 6475 16218 9917 12041 17480 6136 15916 3301 16563 ...
 $ Flight_Status    : chr   "On Time" "Delayed" "On Time" "Cancelled" ...
> colnames(df)
 [1] "Flight_ID"      "Airline"      "Source_City"  "Destination_City"
 [5] "Flight_Distance_km" "Scheduled_Departure" "Departure_Delay_Min" "Arrival_Delay_Min"
 [9] "Ticket_Price"   "Flight_Status"
> df$delay_before <- df$Arrival_Delay
> df$delay_after <- df$Arrival_Delay - rnorm(nrow(df), mean = 5, sd = 2)
> t_test_paired <- t.test(
+   df$delay_before,
+   df$delay_after,
+   paired = TRUE
+ )
> print(t_test_paired)

Paired t-test

data:  df$delay_before and df$delay_after
t = 206.02, df = 6999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 4.942949 5.037920
sample estimates:
mean difference
 5.000435
```